End Semester Examination, April – May, 2025

Degree: **B.Tech**, Semester : **Second**

Stream: **Biotech/CSE(AI)/CSE(AI & ML)/CSE(AIML)/ CSE(Cyber Security) /CSE(Data Science)/CSE(IOT) /CSE(IOT, CS, BT)/ CSE(IOTCSBT)/CSE(Networks) /Robotics and AI /CSE/CSIT/CST/IT/CSBS/ECE/ME/EE/EEE**

Paper Code: **ESCCS201**

Paper Name: **Programming for Problem Solving**

Full Marks: **100** Duration: **3 Hours**

Part - A
Attempt any 10 questions out of 15 questions
Each question carries 2 Marks (2 × 10)

1. Write the output of the following program. **2**

```
#include<stdio.h>
int main(){
  int x = 100;
  printf ("n%d", 1 + (x++));
  return 0;
}
```

2. What is used to exit from a `for`-loop? **2**

3. How do you create an infinite `while`-loop? **2**

4. Write the output of the following program. **2**

```
#include<stdio.h>
int main(){
  int n[3][3] = {
    2, 4, 3,
    6, 8, 5,
```

```
    3, 5, 1};
  printf("%d %d %d\n", *n, n[3][3], n[2][2]);
  return 0;
}
```

5. Write the names of two built-in functions in C. **2**

6. State the difference between an array of pointers and a pointer to an array. **2**

7. Point out the errors, if any, in the following program: **2**

```
#include <stdio.h>
#include <string.h>
int main(){
  struct employee {
    char name[30];
    int age;
    float salary;
  };
  struct employee e;
  strcpy(e.name, "Rabi");
  age = 30;
  salary = 27500.00;
  printf("%s: %d %f\n", e.name, age, salary);
  return 0;
}
```

8. Point out the errors, if any, in the following C program: **2**

```
#include <stdio.h>
#include <stdlib.h>
int main() {
  FILE *fp;
  int c;
  fp = fopen("in.txt", "r");
  if (fp == NULL) {
    puts("Cannot open file\n");
    exit(0);
  }
  while ((c = getc(fp)) != EOF) {
    putchar(c);
  }
  fclose(fp);
  return 0;
}
```

9. When using the statement: **2**

```
fp = fopen("in.txt". "wb");
```

what happens if:

i) `in.txt` is not present in the directory?

ii) `in.txt` is present in the directory?

10. What is `puts()`? Write the syntax. **2**

11. What is the base address of an array? **2**

12. Write the output of the following code: **2**

```
#include <stdio.h>
int main() {
  int a = 19, b = 47;
  a <<= 1;
  b <<= 2;
  printf("Answer = %d.\n", a + b);
  return 0;
}
```

13. Write the output of the following code: **2**

```
#include <stdio.h>
int main() {
  for (int i = 0; i <=2; i++)
    printf("%c %c---", 'F' + i, 'R' + i);
  return 0;
}
```

14. Write the output for the following piece of code. **2**

```
int n = 3, *p;
p = &n; *p = n*n; n = *p * *p;
printf("%d, %d", *p, n);
```

15. Which of the following is/are not functions of `string.h`? **2**

`strlen, strcpy, strman, strncmp.`

Part - B

Attempt any 6 questions out of 9 questions

Each question carries 5 Marks (5 × 6)

16. Write a C program to delete the last two digits of a given positive integer. For **5**

example, if the input is 19324, then the output is 193.

17. Write a C program to find the largest among three real numbers. Assume that the three numbers are distinct. **5**

18. Write a C program using a `for`-loop to calculate the summation of: `1 + 1/(2*2) + 1/(3*3) +...+ 1/(n*n)`, where `n` is the input. **5**

19. Write a C program to compute the summation of three numbers using a function. **5**

20. Write the names of any three header files in C and state a function associated with each of these header files. **5**

21. Differentiate between structure and union in C. **5**

22. Given the following structure and declaration, write a C function to add two complex numbers `a` and `b`. **5**

```
typedef struct {
  float real, float imaginary;
} complex;
complex a, b;
```

23. Given a positive integer as input, write a recursive function to print its unsigned binary representation. Assume that we have 32-bit words to represent integers. **5**

24. What are the best, worst, and average-case time complexities for inserting an element into a sorted array? **5**
Explain in brief. Assume that binary search is used to find the insertion position.

Part - C
Attempt any 5 questions out of 8 questions
Each question carries 10 Marks (10 × 5)

25. Write a C program to sort an input array of integers using Bubble Sort. **10**

26. Write a C program to print the elements of a 1D array in reverse order. **10**

27. Write a C program to efficiently search for a specific key in a sorted array and print its index. If the search key is not present in the array, return -1. **10**

28. Write a C program to count the number of lines in a given text file. **10**

Assume that there is no blank line, i.e., each line contains one or more characters, and only the last line is a blank line.

29. Write a C program to store details of multiple persons (structure with name and age) and find the oldest among them. **10**

30. Write a C program that takes as input a positive integer `n` and prints all primes less than `n+1`. **10**

31. Given a matrix M with integer elements, write a C program to compute and print the sum of elements of every row. **10**

32. An alphanumeric string `x` is said to be *pseudo-palindrome* if removal of exactly one character makes `x` a palindrome. Write a C program to check whether a given string is a pseudo-palindrome. **10**

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